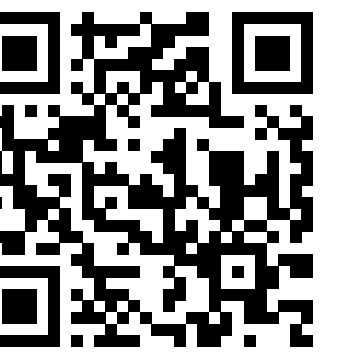
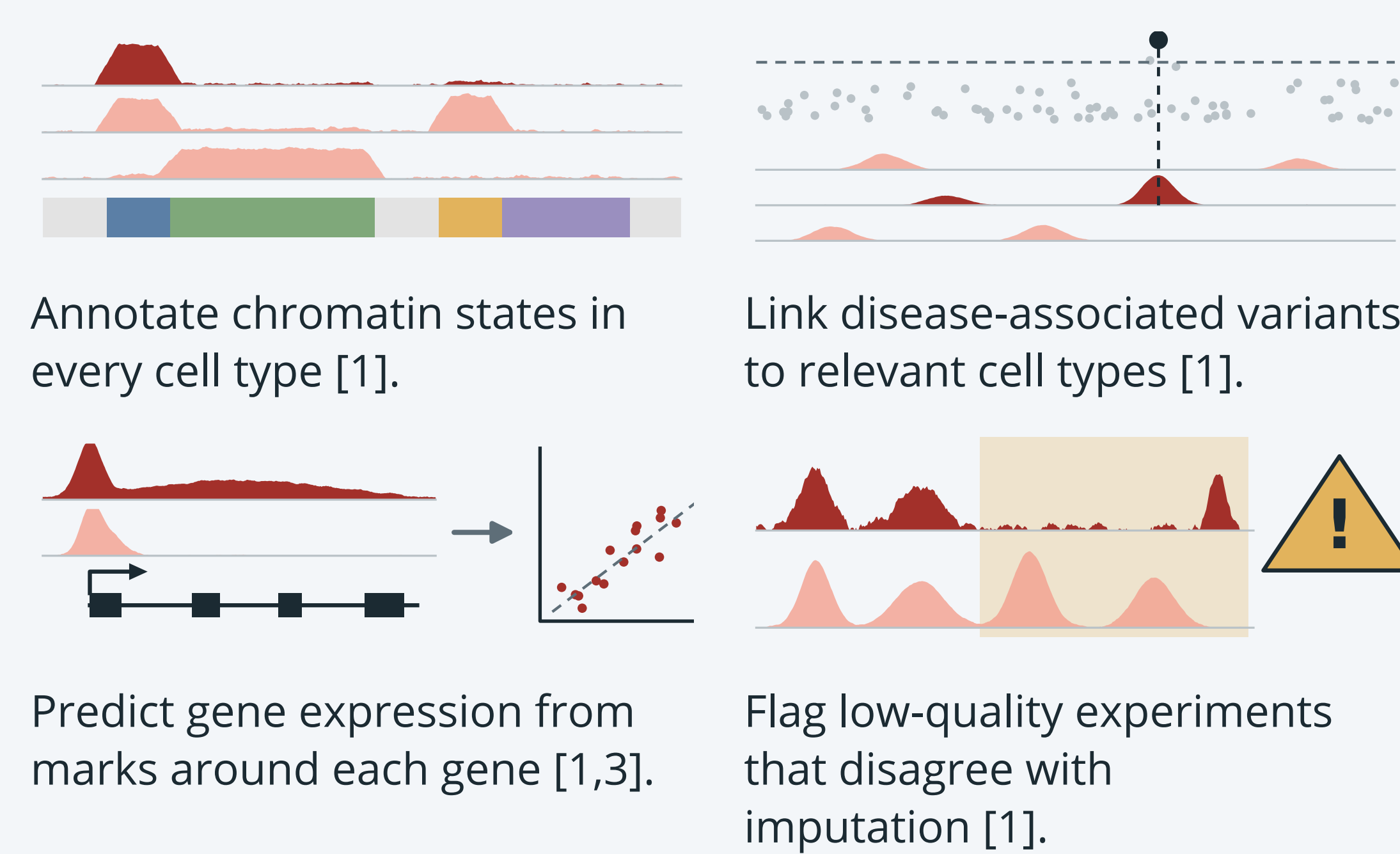




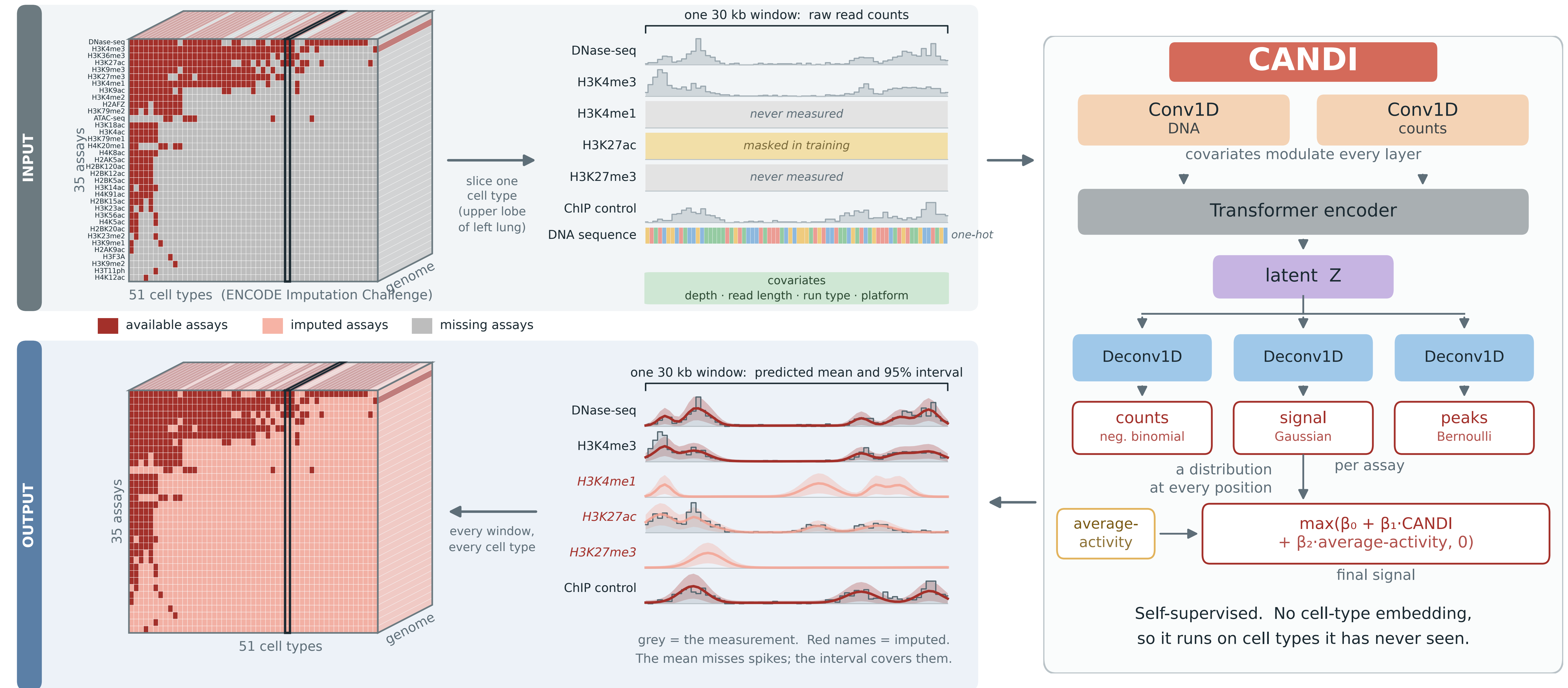
The problem

- Epigenomic data form a 3-D tensor: tens to hundreds of assays $\times$ thousands of cell types $\times$ genomic positions.
- Profiling every assay in every cell type costs too much, so most experiments are missing.
- Imputation predicts these missing experiments [1,2].
- Imputed data are often cleaner than observed data, so imputation also denoises performed experiments [1].
- Experimental covariates matter: in the ENCODE Imputation Challenge, a train-test shift let a simple average-activity baseline beat most entrants [2].

Downstream applications

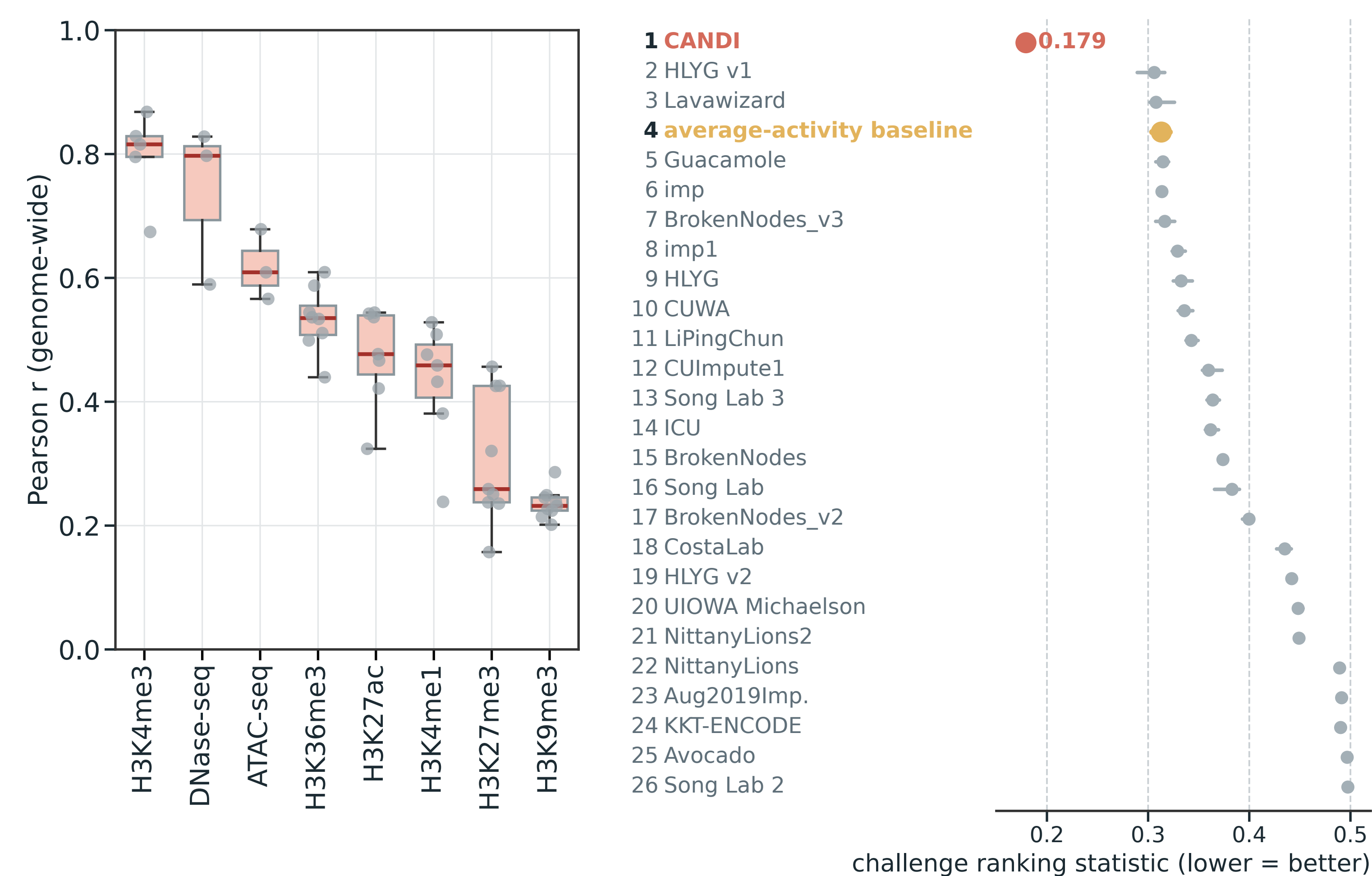


CANDI schematic workflow



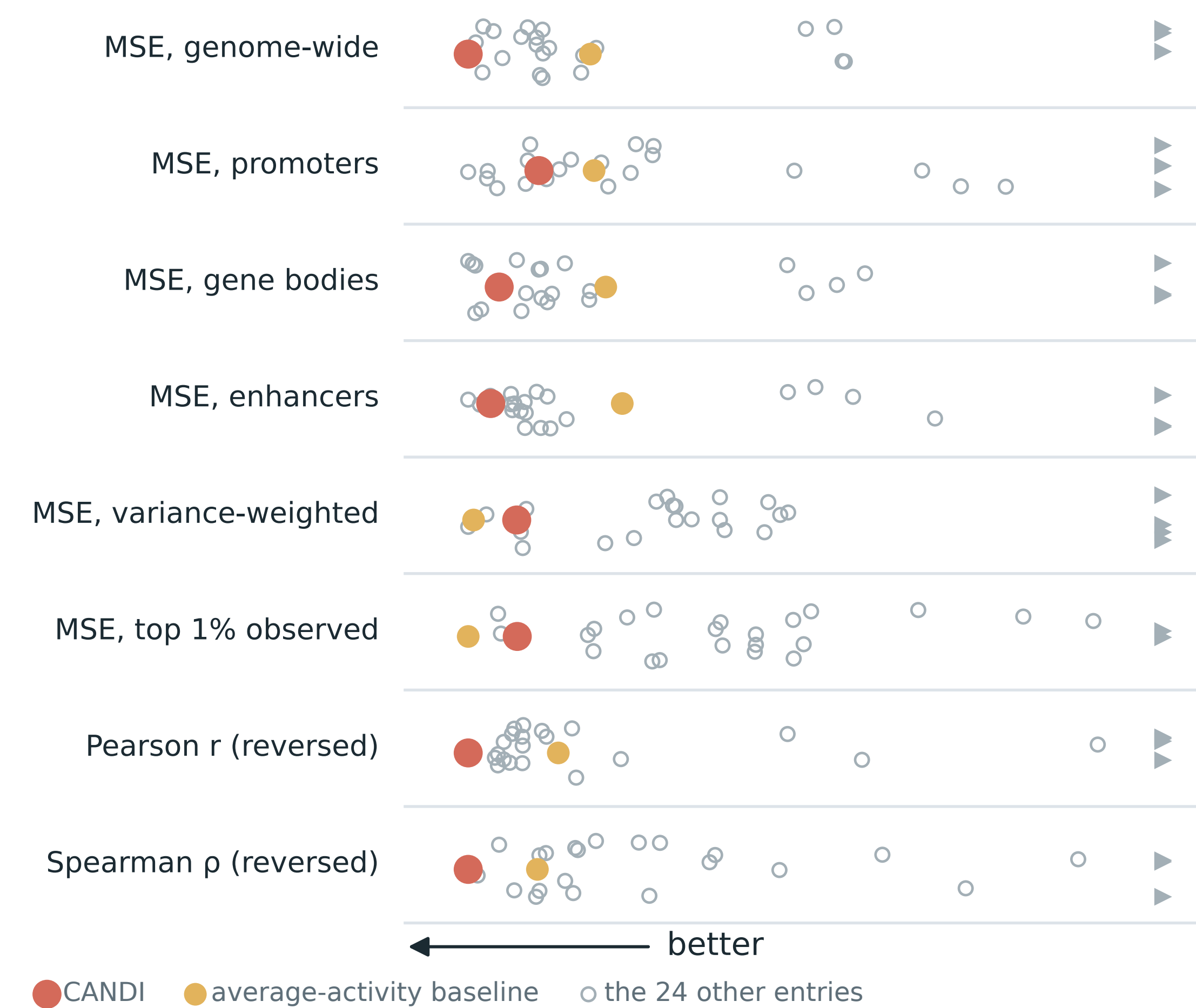
Top: for each 30 kb locus, CANDI takes observed read counts with the ChIP-seq control, four covariates per assay and the DNA sequence; grey assays were never measured, yellow ones are masked. **Bottom:** each output is a probability distribution, here a mean and 95% interval per assay; window by window, they fill every missing thread of the tensor.

Fixing the train-test experimental covariate shift in the ENCODE Imputation Challenge



A Genome-wide Pearson r of CANDI against each of the 51 blind-test experiments (points), by assay.

B The leaderboard with CANDI added: CANDI is first at 0.179 (bootstrap range 0.177–0.184), the next entry 0.306.



C Eight of the nine measures relative to the baseline, median over 51 experiments. CANDI is first on three and top 8 on the rest.

Downstream utility of CANDI

